Joonsuk Bae

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Experimental high-energy nuclear physicist building methodological infrastructure for precision hard-probe measurements. Main analyzer of the first ALICE Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ *TeV (preliminary; paper in preparation; EPS-HEP 2025 PWG-JE merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.*

Education

Ph.D. Candidate, Physics, Sungkyunkwan University — 2022.03 – expected 2027.02
Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at* $\sqrt{s} = 13.6$ *TeV with ALICE.* Defense expected 2026.12; degree conferral 2027.02.

B.S., Physics, Sungkyunkwan University — 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN — 2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory — 2024.02 – present
LAMPS Collaboration, RAON — 2022.03 – 2024.04

Research Highlights

- **Main analyzer**, first ALICE Run 3 inclusive charged-particle jet cross section at $\sqrt{s} = 13.6$ TeV: responsible for the full chain (O^2Physics task, QA, unfolding, and a systematic-uncertainty program carried down to the track level); PWG-JE-approved preliminary; presented on behalf of ALICE at EPS-HEP 2025 (PWG-JE merge talk).
- **PWG-JE tracking-efficiency systematics for Hard Probes 2026**: contributed the pp charged-particle tracking-efficiency systematics at $\sqrt{s} = 13.6$ and 5.36 TeV (p_{T}-binned, 2022/2023/2024 periods); ITS-TPC matching code reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **Central-framework contribution to ALICE O^2Physics** (AliceO2Group/O2Physics, 2023 – present): authored the PWG-JE cross-section normalization framework used by the SKKU Run 3 charged-jet analyses (pp 13.6 TeV and OO — the latter led by two MSc co-analyzers I mentor), and merged a central-framework fix to the track momentum-resolution smearing methodology shared across analyses.
- **ePIC Barrel Imaging Calorimeter**: lead analyst for the CERN PS T10 (2025.07) Pb/scintillating-fiber campaign; DAQ and prompt analysis at KEK AF–AR (2025.03); co-author on the 2024.08 campaign paper (submitted, NIMA-D-26-00482; arXiv:2604.22647), with module QA and assembly.

Publications

Full list at inspirehep.net/authors/2117232; ALICE Collaboration co-author since 2022.10.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE* (submission expected 2026). *[main analyzer]*
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, "Beam test of a Pb/SciFi prototype for the BIC at the EIC," submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; arXiv:2604.22647). *[co-author; module support and PMT/module performance characterization]*
- Y. J. Kim *et al.* (LAMPS), "Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON," *J. Korean Phys. Soc.* **87** (2025) 670–682. *[co-author; Time-of-Flight maintenance and on-site checks]*

Software and Computing

ALICE O^2Physics (AliceO2Group/O2Physics, 2023 – present): 37 PRs, 18 merged. Key contributions:

- **Central-framework fix** to the track momentum-resolution smearing methodology shared across analyses (merged).
- **Event-generator production for model comparison**: managed multi-setup POWHEG-BOX and PYTHIA 8 productions (Monash, colour-reconnection modes, rope and shoving hadronization) plus HERWIG 7 samples for the Run 3 jet-radius model comparison, with provenance tracking across generator configurations.
- **Authored the PWG-JE cross-section normalization framework** used by the SKKU Run 3 charged-jet analyses for pp 13.6 TeV and O+O (merged).

Invited and Contributed Talks (since 2022)

Summary: 3 invited workshops · 5 invited collaboration-meeting talks · 8 international conference contributions (including the EPS-HEP 2025 merge talk on behalf of ALICE and a Hard Probes 2026 parallel talk) · 9 domestic (KPS) · regular PWG-JE working-group speaker since 2023.

Invited (external workshops, seminars).

- **FKPPN Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **HIM 2025**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Invited collaboration-meeting talks.

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed talks and posters.

- **Hard Probes 2026**, Nashville *[international, talk]* — *Jet-like dihadron correlations in pO, OO, and Ne–Ne collisions with ALICE* — 2026.06
- **Hard Probes 2026**, Nashville *[international, poster]* — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions* — 2026.06
- **EPS-HEP 2025**, Marseille *[on behalf of the ALICE Collaboration; PWG-JE merge talk]* — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.07
- **INPC 2025**, Daejeon *[international, talk]* — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE* — 2025.05
- **Quark Matter 2025**, Frankfurt *[international, poster]* — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **Hard Probes 2024**, Nagasaki *[international, poster]* — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2024.09
- **ATHIC 2023**, Incheon *[international, poster]* — *Dijet studies at the LHC* — 2023.04
- **Korean Physical Society (KPS) Spring and Fall meetings** *[domestic]* — 9 contributed talks and posters across Spring and Fall meetings, 2022.10 – 2026.04 (every meeting since joining ALICE).

Detector and Test-beam Experience

- **ePIC Barrel Imaging Calorimeter** — three beam-test campaigns — **CERN PS T10** (2025.07; lead analyst), **KEK AF–AR** (2025.03; DAQ, equalization, prompt analysis), and **CERN PS T10** (2024.08; module performance characterization); ran the PMT-module QA for the prototype modules and contributed Pb/scintillating-fiber bundle assembly; JANA2 / EIC-recon studies for sampling-fraction and energy response.
- **ALICE ITS2 alignment monitoring** (CTF level) — service work providing period-by-period tracking-performance diagnostics from compressed time-frame data, below the standard AO2D analyzer interface.
- **LAMPS Forward Tracker** (2022 – 2024) — MPPC/scintillator-plane forward-tracking concept design and Geant4 feasibility simulations.

Service and Mentoring

- **Paper Committee member**, ALICE PWG-JE — OO charged-particle jet measurement led by W. Ham (SKKU).
- Mentor of two MSc co-analyzers on the ALICE Run 3 OO charged-jet measurement — W. Ham (on jet reconstruction, unfolding, normalization, systematic uncertainties, and model comparisons) and one further co-mentee; co-author with W. Ham on the corresponding PWG-JE Analysis Note.
- Co-organizer and instructor of the **KoALICE $\mathbf{O^2}$Physics Tutorial** (SKKU, 2024.01.22–23; $\sim$22 participants from the Korean ALICE community; co-organized with H. Lee).
- ePIC Korea-BIC analysis liaison between the SKKU group and the Korea-BIC group.

Technical Skills

Programming. C++, Python, ROOT, Geant4, JANA2 / EIC-recon, FastJet, Bash, Git, LaTeX.
Event generators. PYTHIA 8 (Monash, CR modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.
Analysis. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.
Languages. English (professional working proficiency), Korean (native).

References

Available on request.